



DECISION GOVERNANCE INFRASTRUCTURE

UNITED STATES PATENT AND TRADEMARK OFFICE
PROVISIONAL PATENT APPLICATION

TITLE OF INVENTION:

GOVERNANCE CONTROL ARCHITECTURE FOR AUTOMATED DECISION
SYSTEMS WITH SEQUENTIAL CHECKPOINT PIPELINE, DECISION
CONTRADICTION INDEX, AND EXECUTION BOUNDARY ENFORCEMENT

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Inventor:	Harold Alberto Nunes Rodelo
Applicant:	OMNIX QUANTUM LTD (United Kingdom)
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This document constitutes a Provisional Patent Application filed pursuant to 35 U.S.C. § 111(b). It establishes a priority date and grants twelve (12) months from the filing date to submit a corresponding non-provisional application. CONFIDENTIAL — NOT FOR DISTRIBUTION.

FIELD OF THE INVENTION

The present invention relates to computer-implemented systems and methods for governing high-stakes automated decisions. More specifically, the invention relates to a domain-agnostic governance engine that enforces decision admissibility through a sequential pipeline of independent, veto-authority checkpoints; a counterfactual shadow portfolio system for continuous self-calibration of risk management filters; and a Decision Contradiction Index (DCI) for detecting internal signal divergence prior to execution commitment.

BACKGROUND OF THE INVENTION

Automated decision systems operating in high-stakes environments — financial trading, insurance underwriting, medical AI, energy grid management — are capable of generating and executing large volumes of consequential decisions at speeds that preclude human review. The governance of such systems presents a fundamental engineering challenge: how to enforce admissibility requirements at the precise moment a proposed action would transition from a recommendation into an irreversible real-world commitment.

Critical Incidents Illustrating the Governance Gap

Knight Capital Group (August 1, 2012): Knight Capital lost approximately \$440 million USD in 45 minutes after a legacy software module generated thousands of unintended orders for 150 stocks. No architectural mechanism could detect the divergence between intended and actual system behavior at the point of execution. The loss effectively bankrupted the firm.

Flash Crash (May 6, 2010): The U.S. equity markets experienced a rapid 1,000-point drop (≈9%) in minutes. Automated trading algorithms responded in mutually reinforcing patterns with no governance layer capable of detecting the systemic coherence failure.

Zillow Offers Algorithm (2021): Zillow ceased its iBuying program after its price prediction system accumulated overvalued properties, resulting in a \$304 million write-down and 25% workforce reduction. The system continued to make commitments contradicting available market signals, with no checkpoint capable of detecting and blocking the divergence.

Deficiencies of Prior Art

Post-Hoc Audit Systems: Record decisions for subsequent review — do not prevent harm, only create records after harm has occurred.

Single-Layer Risk Limits: Apply a single threshold check — structurally fragile; if the limit check is incorrect or corrupted, no alternative enforcement layer exists.

Consensus Aggregation: Aggregate signals into a composite score, masking internal signal disagreement by design — a fatal flaw when signals strongly contradict.

Static Threshold Systems: Apply fixed thresholds calibrated at deployment; do not adapt to changed market conditions.

Human Review Queues: Incompatible with high-frequency automated environments where review latency is operationally unacceptable.

SUMMARY OF THE INVENTION

The present invention provides a multi-checkpoint governance architecture that enforces decision admissibility at the execution boundary through five integrated components:

Comp onent A	Sequential Checkpoint Pipeline (CP-0 through CP-11)	A pipeline of twelve independent, sequentially-ordered checkpoints, each with independent veto authority. A BLOCK verdict from any single checkpoint prevents execution regardless of the verdicts of all other checkpoints. Default verdict on error: BLOCK (fail-closed).
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Comp onent B	Decision Contradiction Index (DCI)	A real-time signal divergence detector that computes pairwise divergence across a minimum of five independent signal sources. A DCI score of 70–100 (CONTRADICTIONARY classification) constitutes a mandatory BLOCK regardless of individual signal values.
Comp onent C	Counterfactual Shadow Portfolio	A parallel shadow system that receives every proposed action — approved and blocked — enabling continuous calibration of governance threshold effectiveness without survivorship bias.
Comp onent D	Trajectory Invariant Enforcement (TIE)	A post-execution behavioral trajectory audit layer that verifies the resulting system state is consistent with the system's established behavioral trajectory. Triggers circuit-breaker protocols on inadmissible deviation.
Comp onent E	Cryptographic Decision Trace	A post-quantum cryptographically sealed (Dilithium-3 / ML-DSA-65) record of every checkpoint evaluation, verdict, and final admissibility determination.

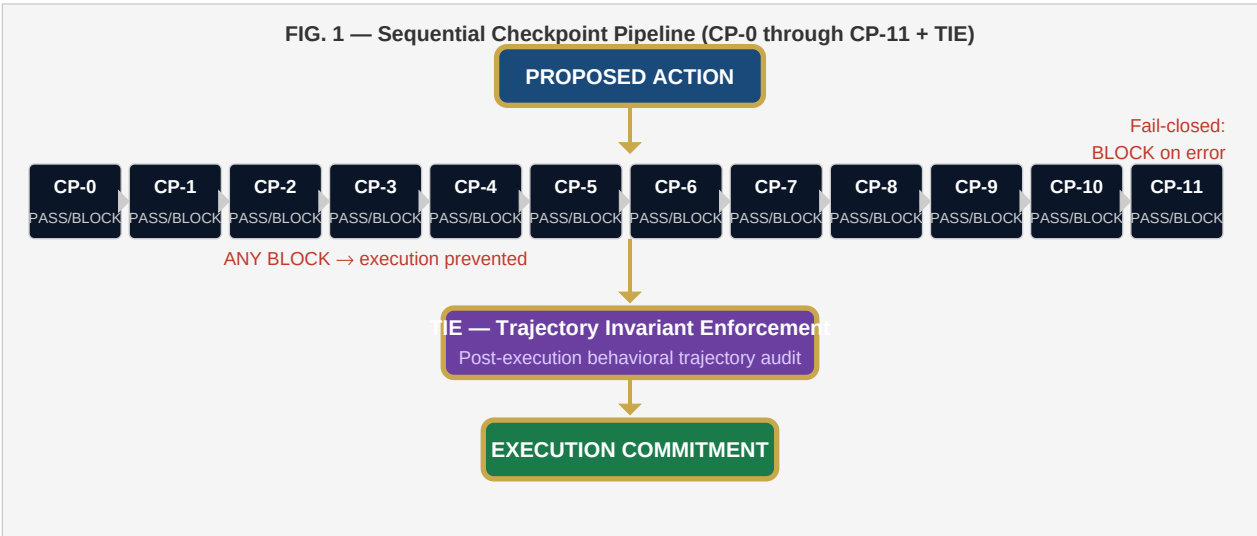


FIG. 1 — Sequential Checkpoint Pipeline with Fail-Closed TIE Post-Execution Audit

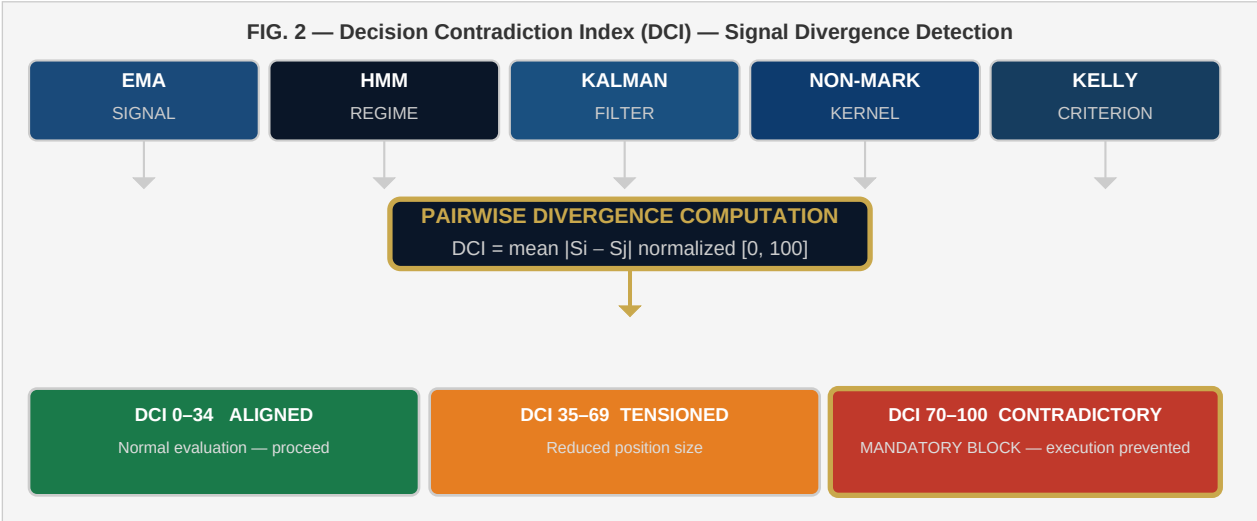


FIG. 2 — Decision Contradiction Index: 5-Source Pairwise Divergence → Three Classification Bands

DETAILED DESCRIPTION OF THE INVENTION

I. Execution Boundary Principle

The System is designed to operate at the execution boundary — the precise computational point at which a proposed action transitions from a computed recommendation into an irreversible real-world commitment. This placement is architecturally fundamental. A proposed action that was valid at time T may be inadmissible at time T+1 if conditions have changed; the System enforces admissibility at T+1, not at T.

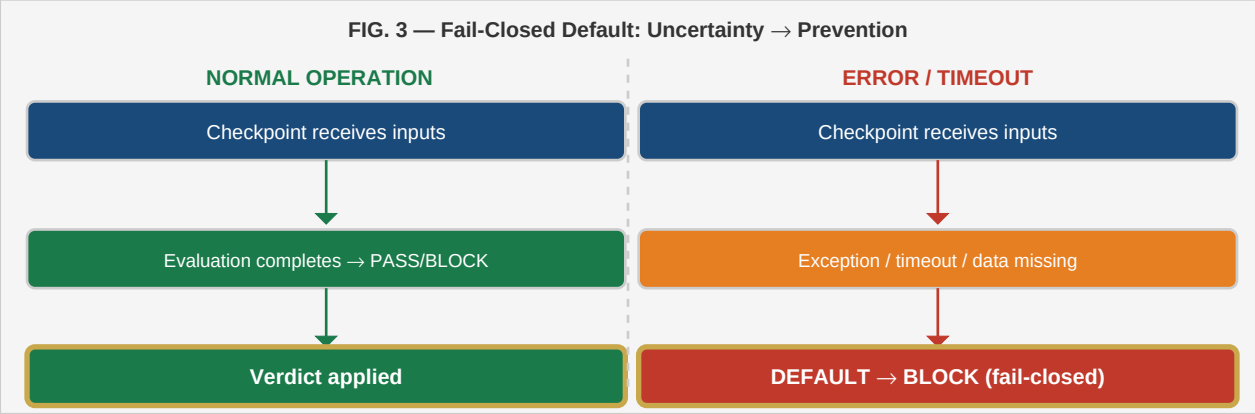


FIG. 3 — Fail-Closed Guarantee: Checkpoint error defaults to BLOCK, never PASS

II. Checkpoint Pipeline Architecture

The sequential checkpoint pipeline comprises twelve checkpoints (CP-0 through CP-11) arranged in a predetermined sequential order. Each checkpoint: (a) operates independently of all other checkpoints; (b) has unilateral veto authority — its BLOCK verdict is individually sufficient to prevent execution; (c) defaults to BLOCK on any evaluation failure (fail-closed); and (d) records its evaluation in the cryptographic decision trace.

CP	Name	Function
CP-0	Signal Completeness	Verifies all required governance signals are present and within freshness thresholds
CP-1	Data Integrity	Verifies cryptographic integrity of all input data using PQC hash verification
CP-2	Regime Classification	Evaluates current market/operational regime classification and confidence
CP-3	Probabilistic Confidence	Verifies composite signal confidence meets minimum threshold
CP-4	Risk Exposure	Evaluates proposed action against current portfolio/system risk exposure limits
CP-5	Volatility Adjustment	Adjusts evaluation thresholds based on current volatility regime
CP-6	DCI Evaluation	Computes Decision Contradiction Index across all signal sources; CONTRADICTION → BLOCK
CP-7	Non-Markovian Memory	Evaluates action against historical memory kernel regime classification
CP-8	Kelly Position Sizing	Verifies proposed action size does not exceed Kelly-criterion-derived maximum
CP-9	Drawdown Protection	Evaluates action against maximum allowable drawdown constraints
CP-10	Correlation Exposure	Evaluates cross-asset or cross-system correlation concentration risk
CP-11	Jurisdiction Compliance	Verifies regulatory admissibility for jurisdiction, asset class, and operation type

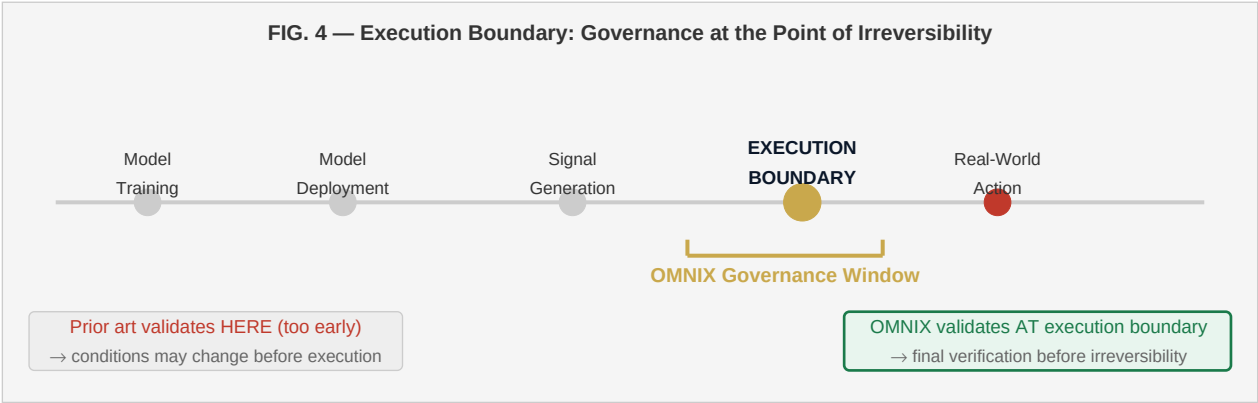


FIG. 4 — Execution Boundary: Prior art validates at training/deployment; OMNIX validates at irreversibility point

III. Decision Contradiction Index (DCI)

The DCI is computed as the mean pairwise absolute divergence across a minimum of five independent signal sources, normalized to the interval [0, 100]. Signal sources include: EMA-based trend signal, Hidden Markov Model regime classifier, Kalman Filter state estimator, Non-Markovian Memory Kernel regime classifier, and Kelly Criterion position signal. The DCI provides a single scalar metric of internal signal coherence:

DCI Range	Classification	Governance Action
0 – 34	ALIGNED	Normal evaluation proceeds through remaining checkpoints
35 – 69	TENSIONED	Position size reduced; evaluation proceeds with heightened scrutiny
70 – 100	CONTRADICTIONARY	MANDATORY BLOCK — execution prevented regardless of other checkpoint verdic

IV. Counterfactual Shadow Portfolio

A shadow portfolio system maintains a parallel record of all proposed actions — both those approved and those blocked — enabling continuous calibration of governance threshold effectiveness. Because both approved and blocked decisions are tracked, the calibration system operates without survivorship bias: it evaluates not only the outcomes of executed decisions but also the counterfactual outcomes of decisions that were blocked. Thresholds are adjusted based on observed filter accuracy over rolling historical windows, subject to mandatory human review before any threshold change.

CLAIMS

What is claimed is:

1. A computer-implemented governance system for automated decision enforcement, comprising: a sequential checkpoint pipeline comprising a plurality of independent checkpoints, each checkpoint configured to evaluate a proposed automated action against assigned admissibility criteria and issue a verdict of PASS or BLOCK; wherein a BLOCK verdict from any single checkpoint is independently sufficient to prevent the proposed action from reaching an execution environment; wherein each checkpoint is configured to issue a BLOCK verdict as a default response when unable to complete its evaluation; and a cryptographic record generator configured to produce a post-quantum cryptographically sealed record of the evaluation prior to any execution commitment.
2. The system of claim 1, wherein the sequential checkpoint pipeline comprises at least twelve checkpoints arranged in a predetermined sequential order, including a signal completeness checkpoint, a data integrity checkpoint, a regime classification checkpoint, a probabilistic confidence checkpoint, a risk exposure checkpoint, a volatility adjustment checkpoint, a Decision Contradiction Index checkpoint, a non-Markovian memory checkpoint, a position sizing checkpoint, a drawdown protection checkpoint, a correlation exposure checkpoint, and a jurisdictional compliance checkpoint.
3. The system of claim 1, further comprising a Decision Contradiction Index (DCI) computation module configured to: receive signals from at least five independent signal sources; compute pairwise absolute divergence across all signal source pairs; produce a normalized DCI score in the range [0, 100]; and issue a mandatory BLOCK verdict when the DCI score meets or exceeds a CONTRADICTORY threshold, wherein the BLOCK verdict is effective regardless of the verdicts of any other checkpoint.
4. The system of claim 3, wherein the at least five independent signal sources comprise an exponential moving average trend signal, a Hidden Markov Model regime classifier, a Kalman Filter state estimator, a Non-Markovian Memory Kernel regime classifier, and a Kelly Criterion position signal.
5. The system of claim 1, further comprising a counterfactual shadow portfolio system configured to maintain a parallel record of all proposed actions including both approved and blocked actions, and to calibrate governance threshold parameters based on the observed outcomes of both approved and blocked actions without survivorship bias.
6. The system of claim 1, further comprising a Trajectory Invariant Enforcement (TIE) layer configured to verify, after execution commitment, that the resulting system state is consistent with the system's established behavioral trajectory; and to trigger circuit-breaker protocols when the resulting state represents an inadmissible deviation.
7. The system of claim 1, wherein the post-quantum cryptographically sealed record comprises a digital signature generated using a lattice-based signature algorithm standardized by NIST under FIPS 204 (ML-DSA), covering all checkpoint evaluations, their verdicts, and the final admissibility determination.
8. A computer-implemented method for governing automated decisions at an execution boundary, comprising: receiving a proposed action from a decision generation subsystem; sequentially evaluating the proposed action through a plurality of independent enforcement units, each having independent veto authority; blocking the proposed action from execution if any enforcement unit issues a BLOCK verdict; defaulting to BLOCK if any enforcement unit is unable to complete its evaluation; and generating a tamper-evident record of the complete evaluation prior to any execution commitment.
9. The method of claim 8, further comprising computing a Decision Contradiction Index by determining pairwise divergence across at least five independent signal sources; and issuing a mandatory BLOCK when the computed index exceeds a CONTRADICTORY threshold.

10. The method of claim 8, wherein the execution boundary is the computational point at which the proposed action would transition from a computed recommendation into an instruction transmitted to an execution environment causing an irreversible real-world action, and wherein all evaluation units complete their evaluations before any such instruction is transmitted.

ABSTRACT

A governance control architecture for automated decision systems enforces decision admissibility at the execution boundary — the precise computational point at which a proposed action becomes irreversible — through a sequential pipeline of independent, fail-closed checkpoints (CP-0 through CP-11), each with unilateral veto authority. A Decision Contradiction Index (DCI) detects internal signal divergence across a minimum of five independent sources and mandatorily blocks execution when pairwise divergence exceeds the CONTRADICTORY threshold. A counterfactual shadow portfolio system enables continuous threshold calibration without survivorship bias by tracking both approved and blocked decisions. A Trajectory Invariant Enforcement (TIE) layer provides post-execution behavioral trajectory audit with circuit-breaker capability. All checkpoint evaluations are sealed in a post-quantum cryptographic decision trace (ML-DSA-65 / Dilithium-3). The architecture is domain-agnostic and applies to financial trading, insurance underwriting, medical AI, energy governance, and autonomous agent systems.

Application Reference: OMNIX-PAT-2026-001 | Inventor: Harold Alberto Nunes Rodelo | OMNIX QUANTUM LTD, United Kingdom | ©
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